

UKRAINIAN AIR-RAID ALERT ANALYSIS

Conclusions, confidence levels & next steps

Official siren dataset • oblast level

Trustworthy window: 15 Mar 2022 – 31 Oct 2025

ONE-LINE VERDICT This data is **excellent for describing and detecting patterns**, and **poor for forecasting the timing of individual alerts**. Both halves are backed by metrics in this report, not asserted.

Colour legend

CONFIDENT

SUGGESTIVE

NOT
CONCLUDABLE

DATA CAUTION

Plain-language summary

We analysed roughly four years of official air-raid alerts for Ukraine, by region (oblast). After cleaning, the period we can reliably study runs from mid-March 2022 to the end of October 2025.

- Alerts follow a clear **daily rhythm**: fewest around 5–7 a.m. local, most around midday, with an evening rise.
- Alerts are **concentrated in front-line and eastern/southern regions**; western regions get far fewer.
- Overall war intensity **rose and fell**, peaking around early-to-mid 2024.
- Some symbolic dates see more alerts — most reliably **New Year**, and to a lesser degree Western Christmas and Women's Day. The **invasion anniversary (24 Feb)** tends to have fewer, not more.
- We **could not predict** when an individual alert will happen — expected, since alerts are adversary-driven.
- We **can reliably flag unusually intense days** after the fact, and describe the patterns above with confidence.

What we are confident about

Large samples (tens of thousands of alerts); stable across the trustworthy window.

CONFIDENT

Diurnal (hour-of-day) seasonality

Robust in Kyiv local time: trough ~05:00–07:00 (~2.2%/hr) rising to a midday peak (~12:00, ~5.4%) plus an evening bump near 21:00. ~2.5× more likely at midday than dawn. Based on ~65,000 alerts.

CONFIDENT

Regional concentration

Donetsk (~6,900), Zaporizhzhia (~6,700), Kharkiv (~6,500), Dnipropetrovsk (~5,600) dominate; western oblasts ~700 each. A structural feature of the conflict's geography.

CONFIDENT

War-phase trend (non-stationarity)
Daily volume is strongly non-stationary: high mid-2022, winter-2022/23 lull, peak around early-mid 2024 (busiest month Mar 2024 ~2,200; busiest day 2022-08-24 with 189), then decline through 2025.

CONFIDENT

Anomaly detection works (descriptive)
STL-residual rolling z-score ($|z| \geq 3$) reliably surfaces real mass-strike days. 37 flagged across all oblasts. This is detection, not prediction.

CONFIDENT

Duration is heavy-tailed
Non-censored durations: median ~40 min, p90 ~182 min, p99 ~540 min. Front-line/some western oblasts higher (~60 min) than central (~33 min).

What is uncertain or inconclusive

Holiday / special-date effects

Only 3–4 occurrences per holiday — directional patterns, not statistically established. Window ± 5 days vs same-year local baseline.

Date	Effect	Read
New Year (1 Jan)	+28%	Robust — elevated every year
Invasion Anniv. (24 Feb)	−11%	Robust — SUPPRESSED every year
Western Christmas (25 Dec)	+19%	Probable, noisier
Women's Day (8 Mar)	+8%	Probable, noisier
Orthodox Christmas (7 Jan)	−7%	No clear effect
Both Easters, 9 May, 24 Aug, 1 Oct, 28 Jun	~0%	No detectable effect

READ HONESTLY New Year (up) is the single most credible holiday finding (survives both ± 3 and ± 5 day windows). The 24 Feb *suppression* contradicts the intuition that anniversaries draw heavier attacks — consistent across years, but only 3 data points. Do not over-claim.

SUGGESTIVE

Forecasting alert timing — conclusively limited
A firm *negative* finding. Daily Kyiv City NB-GLM did not beat naive baselines (MAE ~0.90 vs ~0.87; series mean ~1.3/day). Hourly P(alert) logistic AUC ~0.57 (chance = 0.50), Brier tied with base rate. Onsets are adversary-driven and the diurnal cycle is itself non-stationary.

SUGGESTIVE

Single-region fine-grain seasonality — insufficient signal
The diurnal pattern that is strong nationally is too weak/noisy to exploit at Kyiv-City hourly resolution on held-out data. Aggregate signal is not per-city-hour predictability.

NOT CONCLUDABLE

Luhansk — no data
Only 1 record in the entire official oblast-level set. Excluded everywhere.

NOT CONCLUDABLE

Pre-15 Mar 2022 & post-Oct 2025 — out of scope
Feed starts 2022-03-15 (misses first ~3 weeks) and degrades after Oct 2025. This report says nothing about those periods.

Data-quality issues (read before trusting any number)

Each would have corrupted results if missed.

① **Massive duplication (handled).** ~50% of raw oblast rows were exact duplicates — a publisher-side daily-regeneration artifact (history block written twice). Removed via provably-lossless full-row dedup. Must remain a permanent pipeline step.

② **Feed collapse after Oct 2025 (truncated).** From Dec 2025 only Kyiv City reports; 24 other oblasts go silent. This is a collection failure, NOT a ceasefire. Hard cut at 31 Oct 2025. The late-2025 'drop to zero' is an artifact.

③ **Right-censored durations (flagged, not deleted).** Missed 'all-clear' inflates duration. Only 13 alerts exceed 24h, all in front-line oblasts where long alerts can be real. Excluded from duration stats, kept in counts — so counts are robust, durations use non-censored only.

④ **Timezone & granularity.** Raw is UTC; all hour/day analysis uses Kyiv local (Europe/Kyiv, DST-aware). CSV round-trips drop tz metadata (re-localise downstream). Oblast level only — raion/hromada rows double-count and hold extreme outliers.

Methods & key metrics

Element	Choice & rationale
Primary metric	Alert ONSET COUNTS, not durations — robust to censoring.
Decomposition	STL period=7 (weekly). Annual seasonality weak/irregular.
Anomaly model	STL residual ÷ 30-day rolling local std; flag $ z \geq 3$. Global z rejected (would mislabel busy 2024).
Holiday test	± 5 -day window vs same-year ± 21 -day local baseline. Global-mean pooling invalid (trend confound). $n=3-4$.
Daily forecast	Negative-Binomial GLM (mean~1.3, overdispersed → ARIMA unsuitable) vs mean & seasonal-naive baselines.
Hourly forecast	95% empty hours → model $P(\geq 1 \text{ alert})$ via logistic; score with ROC-AUC & Brier, not point accuracy.
Validation	Strict time-based split (recent 20% held out, no shuffle).

Key numbers. 65,128 clean alerts • 25 regions (24 usable) • window 2022-03-15...2025-10-31. Daily Kyiv City: mean 1.32, var 1.86 (overdispersion 1.41), 32% zero-days. Hourly: 95% zero-hours, base rate ~5.4%. Daily MAE seasonal-naive 0.865 vs NB 0.895. Hourly AUC 0.569; Brier 0.0551 (vs 0.0553 baseline).

How to use this & next steps

USE FOR

- Situational briefing (diurnal curve, regional ranking, trend)
- Retrospective anomaly flagging
- Baselines any real model must beat
- Hypothesis generation (test holiday signals externally)

DO NOT USE FOR

- Predicting individual alert timing
- Reading anything into post-Oct-2025 data
- Duration claims that include censored alerts

Next steps, in priority order

1 • Spatial nowcasting (highest value)

Published research on this dataset found a region's alert state is best predicted by its **neighbours' current state** — alerts sweep geographically. Target $P(\text{region enters alert in next 15–30 min} \mid \text{neighbour states})$. Build a minute-resolution per-oblast 'active' panel; use lagged neighbour states as features. First measure actual cross-region onset lags to confirm the signal exists at the chosen horizon.

2 • External event fusion

Join alert spikes with documented mass-strike / infrastructure-campaign datasets to validate the anomaly detector and test holiday hypotheses with real significance (in-dataset n is too small alone).

3 • Duration / survival modelling

Treat censored alerts properly (survival analysis) rather than excluding them, especially for front-line oblasts where long alerts are real.

4 • Backfill early-war data

Optionally splice the volunteer dataset (from 2022-02-25) to recover the first ~3 weeks, with explicit source-comparison checks.

5 • Hourly model with spatial + recent-activity features

The calendar-only hourly model failed (AUC 0.57). Adding recent national activity and neighbour states (i.e. merging step 1) is the only plausible route to beating the base rate at hourly resolution.

BOTTOM LINE

Definite: diurnal seasonality, regional concentration, war-phase trend, heavy-tailed durations, and that calendar-based timing forecasting does NOT work.

Suggestive: holiday effects — New Year up, 24 Feb down most credible, but all on 3–4 yearly points.

Not concludable: Luhansk, pre-Mar-2022, post-Oct-2025, per-city-hour predictability.

Biggest caveat: trust no number outside 2022-03-15...2025-10-31.

Most promising next step: spatial nowcasting from neighbouring regions.